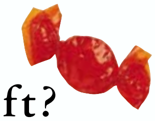
The miraculous jar

Infinity is weird. Imagine a jar containing an infinite amount of sweets.

If you take one out, *how many* are left?



The answer is exactly the same amount: infinity. What if you take out a billion sweets? There'd still be an infinite amount left, so the number wouldn't have changed. In fact, you could take out **half the sweets**, and the number left in the jar ***wouldn't have changed***.

Mathematicians use the symbol ∞ to mean INFINITY, so we can sum up the strange jar of sweets like this:

$$\infty - 1 = \infty$$

$$\infty + 1 = \infty$$

$$\infty - 1{,}000{,}000{,}000 = \infty$$

$$\infty \div 2 = \infty$$

$$\infty \times \infty = \infty$$

But infinity isn't exactly a number – it's really just an idea. And that's why sums involving infinity don't always make sense.

Find out more

Hilbert's Hotel

Mathematician **David Hilbert** thought up an imaginary hotel to show the maths of infinity. Suppose the hotel has an infinite number of rooms but all are full. A guest arrives and asks for a room. The owner thinks for a minute, then asks all the residents to move one room up. The person in room 1 moves to room 2, the person in room 2 moves to room 3, and so on. This leaves a spare room for the new guest.

The next day, an infinitely long coach arrives with an infinite number of new guests. The owner has to think hard, but he cracks the problem again. He asks all guests to double their room number and move to the new number. The residents all end up in rooms with even numbers, leaving an infinite number of odd-numbered rooms free.

Beyond infinity

Strange as it may sound, there are different kinds of infinity, and some are bigger than others. Things you can count, like whole numbers (1, 2, 3 ...), make a ***countable*** infinity. But in between these are endless peculiar numbers like phi and pi, whose decimals places never end. These "irrational numbers" make an ***uncountable*** infinity, which, according to the experts, is infinitely bigger than ordinary infinity. So infinity is bigger than infinity!

walking around planet Earth over and over again. Suppose it takes one footstep every million years. By the time the ant's feet have worn down the Earth to the size of pea, eternity has not even *begun*.